

Cycle Conference

Ohio State University

April 18, 2026

Schedule

10:00-11:00 AM	Speaker check-in, poster set-up, technology check	EA lobby
11:00-11:30 AM	Conference check-in for non-speakers (i.e. parents, professors, etc.)	EA lobby
11:30-1:30 PM	Poster session 1 / Lunch	EA lobby
1:30-4:00 PM	Spoken presentation session (with midpoint break)	EA 160
4:00-5:00 PM	Poster session 2	EA lobby
5:00-6:00 PM	Plenary address by Dr. John Johnson	EA 160
6:00-7:00 PM	Dinner	MW 724
7:00-7:15 PM	Closing	MW 724

Plenary Talk: Dr. John Johnson

Title: The Pigeonhole Principle: Dispatches from the past, present, and (maybe) future

Abstract: Every mathematics student either intuitively knows or quickly learns about the humble Pigeonhole Principle: if you have more pigeons than holes and every pigeon must be placed in a hole, then "clumping" (that is, more than one pigeon in a hole) is inevitable. One often encounters this principle in discrete proofs, but another point-of-view shows that this principle, and its variations, is foundational from basic counting to more sophisticated dynamical arguments where the interplay between uniformity and structure is crucial. In this talk, we will trace the deep history of the Pigeonhole Principle from its humble origins, to sketches of its present use, and hints at the future of this endearing fact.

Contributed Talks

1 William Newman

Title: Polynomial Parameterization

Abstract: A polynomial $f(x_1, \dots, x_n)$ is parameterizable if its zero set can be parameterized by rational functions. We discuss the extensive work on determining which polynomials are parameterizable.

2 Killian Davis

Title: Pathological Triangles

Abstract: Famously, triangles are 3-sided polygons whose interior angles add up to 180 degrees. In this talk, we ask what would happen if the interior angles didn't add up to 180 degrees. This will lead to a discussion of curvature, strange geometry, and the axioms of Euclidean geometry.

3 Daniel Tcheurekdjian

Title: What is Deep Learning Actually?

Abstract: The application of deep learning has yielded significant advancements across domains, from natural language processing to protein discovery. Furthermore, these approaches have several counterintuitive properties: they generalize, approximate arbitrarily complex functions, and converge to similar solutions. It is unclear, however, what deep learning is actually doing in order to achieve these results. One perspective, which we discuss here, is that deep learning is performing program synthesis, that is, it is searching a space of composable programs, guided by the learning objective and by a simplicity bias. We explore this perspective, evaluate its explanatory power on observed phenomena, and highlight what remains to be demonstrated for the perspective to be formalized into theory.

4 Santiago Rodríguez Sierra

Title: Is this your card?

Abstract: Take a break from the conference to come and watch us perform an impossible looking card trick. Then we'll break magic's first rule and reveal exactly how it works. Along the way, you'll discover that the real secret isn't sleight of hand at all, it's some surprisingly clever mathematics.

5 Nathan Willey

Title: Computers Don't Understand Symmetry

Abstract: Humans are naturally good at understanding symmetry in the world around us, but computers don't share this skill. Data is often stored only as collections of numbers. This talk explores how the math of "groups" can help computers understand data's innate symmetry.

6 Gradmar Maldonado Marti

Title: Counterintuitiveness of Infinity: Hilbert's Hotel

Abstract: The concept of infinity is tied to the idea of a never ending procedure. In mathematics, infinite sets are collections with a never ending amount of elements (for example, the natural numbers are an infinite set). Hilbert's hotel shows that a hotel with infinitely many rooms at full capacity can still make room to accommodate more guests (an infinite amount of infinite guests even!), which tells us just how counterintuitive the notion of infinity can be.

7 Hugh Dennin

Title: The Cyclic Sieving Phenomenon

Abstract: Numbers are pretty good at counting. However, often we have a collection X containing objects of various "sizes." In such cases, we may want to count X "multi-dimensionally" using a vector of numbers, which we often write as the coefficients of a polynomial $f(q)$ (called a q -analog of X) whose evaluation $f(1)$ gives the usual count $|X|$. What can be said about $f(-1)$? $f(i)$? A surprising phenomenon is that evaluating $f(q)$ at an n th root of unity often has an enumerative interpretation when the objects of X possess an n -fold symmetry.

8 Kacey Aurum

Title: A Nonsense Introduction to Quantum Gravity

Abstract: What the **** is quantum gravity? We will explore several areas of physics to understand in what situation a quantum gravity theory is needed and discuss two of the most well-studied candidate theories.

Poster Session

1 Sam Allen

Mentor: Tyler Genao

Title: Uniform Bounds on the Level of Cyclotomic Division Fields of Elliptic Curves

Abstract: In this project we proved that for each number field F there exists a uniform bound on the prime levels p of elliptic curves E/F for which $F(E[p]) = F(\zeta_p)$. Under the Generalized Riemann Hypothesis, we also gave uniform bounds on p for which $F(E[p])/F$ is abelian, provided that F has no rational complex multiplication. These are generalizations of results of González-Jiménez and Lozano-Robledo to general number fields.

2 Qichang An, Isaac Choo, Tony Tran

Mentor: Torey Hilbert

Title: Modeling Diffusion with Random Walks: Mesh Parameterization

Abstract: Diffusion and random walks are connected to the Laplacian in many ways. We walk through the derivation of the connection starting with simple random walks on $\mathbb{Z}$ and extending to higher dimensions via carefully specifying the transition matrices. and give some applications.

3 Jackson Bean, Paarth Mahajan

Mentor: Levi Keck

Title: Symplectic Integrators

Abstract: When simulating physical systems, there are a number of choices as to which numerical methods one employs. One such class of numerical methods are known as symplectic integrators. These integrators have certain area preservation properties in the phase space of the system that allow for near total conservation of the total energy, or the Hamiltonian associated with the system. This property arises as a result of the methods being Hamiltonian themselves and solving a modified differential equation with an associated modified Hamiltonian. These properties allow for such numerical methods to be especially ideal for purposes such as molecular dynamics, where energy conservation is important to achieving realistic simulations.

4 Michelle Cai, Lexi (Miaomiao) Cheng

Mentor: Jiashan He

Title: Designs for Clinical Trials with Two Treatment Arms

Abstract: This project is about single-stage and two-stage designs for randomized phase II trials with two experimental arms to identify treatments that increase response rates over a historical baseline. The methodology accurately controls the probability of erroneously selecting an inferior treatment while ensuring the power to correctly identify a superior arm.

5 Wenqi Cai

Mentor: Luci Krnic

Title: Topological Data Analysis of Time series via Taken's Embedding

Abstract: We emulate the paper which uses topological data analysis to process wearable device heart rate data and feeds it as a prior to a machine learning classification algorithm. Instead, we run the topological data analysis pipeline and learn about Taken's embedding to interpret results.

6 Ziyue Chen, Jin Qian

Mentor: Dr. Michael Tychonievich

Title: Minimum-Variance Portfolio with Feasible Set Constraints

Abstract: This project studies a Flexible Investment Plan using mathematical optimization. We examine how to allocate wealth across VOO, GLDM, and BND to minimize portfolio risk. In this model, flexibility does not mean market timing. Instead, it means allowing portfolio weights to adjust within a set of mathematical constraints as risk changes overtime. Using historical return data, we estimate the covariance matrix of asset returns and solve a minimum-variance portfolio problem. We then compare how a fixed allocation and a flexible risk-based allocation perform under the same framework.

7 Julia Circele

Mentor: Giovanni Ferrer

Title: Calculus Y: Why not do Hilbert objects in any category?

Abstract: We define a generalized notion of duality to account for infinite dimensional spaces. Utilizing the graphical calculus for tensor categories, we implement this duality formalism to define Hilbert “spaces” internal to tensor categories. Our main desideratum for these generalized objects is the fundamental theorem of Hilbert spaces: the Riesz Representation Theorem.

8 Gavin Crim

Mentor: Killian Davis

Title: How to Turn Knots into Polynomials

Abstract: A knot is an embedding of a circle S^1 into $\mathbb{R}^3$. Through studying knot invariants, knot theory investigates the geometry of knots and how to distinguish one knot from another. In this project, we investigate ways to turn knots into polynomials in an invariant way. Specifically, we look at the Jones polynomial.

9 Bessie Cui, Aiden Gavin, Nishal Nepal, Nathan Shaner

Mentor: Gradmar Maldonado Marti

Title: Set Theory and Its Philosophy

Abstract: This project focused on reading select topics of Michael Potter’s Set Theory and its Philosophy. We explored the axioms of ZFC and constructed a model of it. We read about the hierarchy of sets and the nature in which the universe of sets behaves. We also explored how to instantiate other mathematics within the universe of sets, in particular we saw the construction of the natural numbers. We saw some basic cardinality facts; the most important being Cantor’s theorem, which establishes that the cardinality of a set is strictly less than the cardinality of its power set.

10 Sandra Fu

Mentor: Luci Krnic

Title: The Shape of a Roster: Persistent Homology Across Four Professional Sports

Abstract: We use persistent homology to analyze sports roster data. We extend a paper which applies analysis to 2014 hockey data to 2025/26 NHL, NFL, baseball and basketball data.

11 **Jiayi Gao**

Mentor: Andrew Gothard

Title: Modeling COVID-19 Spread Using SIR Models

Abstract: Classical infectious disease models use differential equations to model rates of infection in population-based models. We used Dynamic Survival Analysis to extend to agent-based models and allow for incorporating ideas from survival analysis into our estimations. Finally, we estimate parameters for the COVID-19 pandemic in China using Dynamic Survival Analysis.

12 **Nathen Hale**

Mentor: Luci Krnic

Title: Topological Data Analysis of EEG Brain Networks in ADHD and Neurotypical Children

Abstract: Investigation of EEG brain networks of ADHD children compared to typically developing children via a topological data analysis method called persistent homology. Using persistent homology we study the shape of the brain networks by comparing the number of holes and the robustness of these holes.

13 **Michelle Huang, Wanning Rong, Yujia Feng**

Mentor: Jacob Scherzer, Nathan Willey

Title: P3D

Abstract: The goal of our project was to create P3D, an interactive 3D graphing program. To achieve this, we studied concepts from numerical analysis, calculus, and linear algebra to support a wide range of plotting modes. We also developed our programming skills in Python, using the Plotly and Dash libraries to build interactive web-based visualizations. This project highlights a mix between mathematical theory and coding to model functions in three dimensions and make their visualizations easy to explore.

14 **Xuan Huang**

Mentor: Dr. Rima Chatterjee

Title: Knots and the Jones Polynomial

Abstract: We examine the Jones polynomial, a powerful knot invariant, in the context of knot theory. We look at its theory and computation, along with its invariant ability to distinguish knots and links.

15 **Yihan Jiang**

Mentor: Natalie Patten

Title: Classification of Integers via Tunnell's Theorem

Abstract: The congruent number problem asks which positive integers arise as the area of a right triangle with rational side lengths. Despite its simple formulation, determining whether a number is congruent remains highly nontrivial. In this project, we study congruent numbers through a computational application of Tunnell's Theorem.

16 **Sean Lipton**

Mentor: William Newman

Title: Classifying Quadratic Endomorphisms of Galois Planes

Abstract: Providing colorful intuition for ramification and Galois planes with clear examples. We culminate in an unexpected finding that the images of these maps are often fully described by just a few pieces of information.

17 **Ryan Liu, Pranay Boddeda, Haixin Lin**

Mentor: Dr. Connor Cassady

Title: p -adic Numbers

Abstract: The goal of the project was to understand the construction and nature of the p -adic number system, which then expanded into an exploration of potential applications of $\mathbb{Q}_p$ in solutions to polynomial forms.

18 **Andrew Lovelass, Owen Rusnak**

Mentor: Hunter Handley

Title: Zero-Knowledge Proofs

Abstract: Sometimes you want to show that you know information without giving it away. In the case of having a lock, you would put in the code without anyone seeing and then show that it is unlocked. For more complex scenarios, we need to introduce more complexity. For this, we lean on math and trapdoor functions.

19 **Michael Mayano**

Mentor: Dr. Michael Tychonievich

Title: Alpha Decomposition of Momentum and Mean-Variance Optimization Strategies in Factor-Targeted ETF Portfolios

Abstract: To conduct a quantitative comparative analysis of portfolio construction methodologies. This study utilizes the Fama-French 3-Factor framework to determine how naive diversification, momentum-based rotation, and quadratic optimization alter systematic risk exposure and affect the distance of a portfolio from the theoretical Efficient Frontier.

20 **Mohith Muddasani, Steven Wu**

Mentor: Tyler Genao

Title: Galois Representations of Complex Elliptic Curves

Abstract: In this project, we explore connections between the algebra of Galois representations of elliptic curves, and their complex-analytic structure as tori.

21 **Ishan Patel**

Mentor: YiMing Chen

Title: Beyond Black-Scholes: Jump-Diffusion Pricing for Electricity Markets

Abstract: BSM option pricing fails in electricity markets: prices exhibit mean reversion, sudden spikes, and seasonal patterns violating lognormal assumptions. We calibrate a mean-reverting jump-diffusion model on six years of Nord Pool SYS data (2013–2018), achieving 31% lower pricing error ($p < 0.0001$) and $4.97\times$ higher payoffs on spike days ($p = 0.013$) versus BSM.

Why it matters: Accurate pricing improves hedging, risk management, and valuation in volatile electricity markets.

22 Krish Patel

Mentor: Daniel Tcheurekdjian

Title: Initial Reports on Mechanistic Interpretability of Agentic Fine-Tuning For Computer Use in a VLM

Abstract: Fine-tuning turns a vision-language model into an autonomous computer-use agent, taking it from emitting no valid actions to reliably picking the correct one on held-out samples — so what does that training build inside the network? We introduce a Dedicated-Feature Cross-Layer Transcoder (DF-CLT) that splits base and fine-tuned models into sparse shared, base-exclusive, and fine-tune-exclusive features, and use it to identify the agency circuits fine-tuning builds. Attribution graphs reveal a compact action-selection circuit of $\approx 1,300$ features in layers 21–24 — about 5% of the fine-tune-exclusive partition. Ablating it makes the model $3.8\times$ worse at action selection while random ablations do nothing; transplanting it into the untrained base yields a $2.5\times$ improvement, confirming cross-model transfer of action-specific information. A stable ≈ 21 -feature agency core persists across multi-step tasks. These are early findings, but the picture is suggestive: agentic fine-tuning installs a small, identifiable, partially portable piece of machinery enabling agency.

23 Ridham Patel

Mentor: Xiaoting Han

Title: A Deep Flow Map Learning Framework: Constructing ResNet Surrogates for Reaction-Diffusion PDEs

Abstract: Modeling complex dynamical systems, such as 1D reaction-diffusion equations, traditionally requires computationally intensive numerical solvers. This project details the development of a Flow Map Learning (FML) pipeline, a data-driven approach designed to predict system evolution without the need for continuous traditional integration. We focus on a PDE characterized by a spatially varying reaction term $\lambda(x)$ and 2 external forces $u(x, t), u(t)$, using the Crank-Nicolson method to generate a mathematically pristine ground-truth dataset. This data is utilized to construct and train a Residual Neural Network (ResNet) to learn the discrete flow map $\Phi : y(x, t) \rightarrow y(x, t + \Delta t)$. Rather than focusing solely on final empirical benchmarks, this work emphasizes the methodological framework required to build such a surrogate. By detailing the data generation bottleneck, network architecture, and multi-step training strategies, we establish a robust foundation for utilizing deep learning to bypass computationally heavy numerical integration for complex physical simulations.

24 Niclous Pi

Mentor: Dr. Michael Tychonievich

Title: Does Rebalancing Improve Portfolio Performance? evidence from a 60/40 Stock-Bond Strategy

Abstract: The primary goal of this project is to examine whether periodic rebalancing can improve the overall performance of a fixed-allocation portfolio.

25 Yun Qiao

Mentor: Luci Krnic

Title: Topological Data Analysis of Financial Market Dynamics: Detecting Systemic Risk via Persistent Homology

Abstract: We use Topological Data Analysis to study how the market crash reshaped the hidden structure of stock and crypto markets. Even with a small sample of 20 assets, we observed a clear “topological collapse” as market diversity vanished into synchronized panic. These results prove that selective market sampling can effectively detect early-warning signals of systemic risk without requiring exhaustive full-market data.

26 **Zihe Ren**

Mentor: Dr. Parisa Fatheddin

Title: Applications of Navier Stokes Equation

Abstract: We are studying the theory and applications of the Navier-Stokes equation, which is an important model in fluid dynamics. We have focused on the two main applications of this equation: Hemodynamics and Aerodynamics. Hemodynamics is the study of blood flows to and from the heart and Aerodynamics involves the examination of the design and efficiency of aircrafts. In these areas, Navier-Stokes equation is used to study the blood and air flow, respectively. Examples of applications include studying the pressure and temperature of the blood and using optimization to minimize noise and maximize fuel efficiency of aircrafts.

27 **Cory Shields**

Mentor: Dr. Richard Haburcak

Title: An Introduction to Birational Equivalence and Applications

Abstract: As a guiding example, we use the rational unit circle and rational line to parametrize all Pythagorean triples. From this, we build up to birational equivalence as a way to relate varieties through rational maps that are invertible on dense subsets, keeping the focus on a concrete case while illustrating how the idea extends more broadly.

28 **Ryland Smoot**

Mentor: Hugh Dennin

Title: Combinatorial Identities of Immanants

Abstract: We explored the combinatorial meaning behind certain immanant computations, utilizing various methods from multiple perspectives.

29 **Braden Stewart**

Mentor: Alison Kraniske

Title: Making Stokes' Theorem Clear

Abstract: The goal of our project was to utilize processes of glass making to create a physical model to visualize Stokes' Theorem. Clear and colored glass were used to show the bounded surface as part of a larger, complete surface. Frameworked glass arrows can be moved around the object to visualize normal vectors and vector fields. The material of glass allows for drawing on the surface to demonstrate circulation and curl. Between the physical object, computer rendering, and the traditional question, we are hoping to increase understanding of a difficult concept for Calculus 3 students.

30 **Xingji Tao, Vyom Goel**

Mentor: Santiago Rodríguez Sierra

Title: Pascal's Triangle and the Sierpinski's Pattern

Abstract: This project follows a progression from geometry to number theory. We found that the Sierpinski triangle is not only geometric, but also arithmetic, connecting the Sierpinski triangle with Pascal's Triangle considered modulo 2.

31 **Jason Tu**

Mentor: Dr. Greg Hemenway

Title: Dynamical System Exploration: AI for Expanding Iterated Function Systems

Abstract: Thermodynamic formalism provides tools to study the ergodic theory of chaotic dynamical systems. In particular, we wish to pick a distinguished measure that sees all of the chaotic behavior; i.e. a measure of maximal entropy or more generally an equilibrium state. Dr. Hemenway studied the structure of ESs in nonstationary dynamics. During this project, we investigate a relative variational principle for expanding iterated function systems, relating topological entropy and measure-theoretic (metric) entropy in the skew-product setting.

32 **Annie Wang**

Mentor: Jacob Herman

Title: Towards Physically Informed Neural Network Structures with NeuroSim Studio

Abstract: Biological and artificial neurons are quite different in the mechanisms by which they learn to represent information, but both can do so accurately. In this work, we emphasize the feasibility of using biological principles to create artificial neural networks. We present a web app which allows users to paint modular neural structures, simulate how their connectivity develops, and translate this connectivity into weights for a neural network via a single interface.

33 **Ziqing Wang**

Mentor: Dr. Michael Tychonievich

Title: Modeling Natural Gas Prices Using Storage Fundamentals and Seasonal Dynamics

Abstract: This project aims to develop a framework to model and forecast weekly natural gas storage changes and derive trading signals based on storage surprises.

34 **Minxiao Wu**

Mentor: Ishan Bannerjee

Title: The Derivative of Period Mapping: Glimpsing the Intrinsic Essence of Hodge Structures Through an Infinitesimal Deformation

Abstract: In this poster we will discuss the basics of Hodge theory and period maps, particularly in the special case of hyperelliptic curves.

35 **Samuel Younkin**

Mentor: Sean Basa

Title: An Introduction to Category Theory

Abstract: This project aims to serve as an introduction to core concepts in category theory, progressing from exposure to misc. mathematical objects in abstract algebra until the Yoneda lemma. With this introduction established, we seek to find points of conversation with recent category-theoretic, proof-theoretic approaches to semantics-syntax.

36 Julie Zhu

Mentor: Dr. Anjun Ma

Title: NeuroCLIP: Learning a Shared Embedding Space for Multimodal Brain Data

Abstract: OmiCLIP is an image foundation model extended from contrastive language-image pretraining methods that integrates single-cell transcriptomics (scRNA-seq) with histology approaches. However, it is not trained on tissues from patients with neurodegenerative diseases. We fine-tune OmiCLIP and find that our new model has improved performance for cross-modal retrieval and alignment for brain diseases.

37 Xiaocheng Zhu

Title: Ghost

Zachary Frager, Ethan Yuwono

Mentor: Kacey Aurum

Title: Busters

Abstract: Ghost is a turn-based word game where players construct a word letter by letter, with the player who completes the word losing the game. Our goal is to find the list of words that would allow for a perfect player (going second), to follow the optimal path for victory. To do this, we've used two separate approaches to find these solutions: a game approach and an exhaustive approach. Through these two approaches, we're able to create a list of optimal paths that a computer could use to force a win.